A simple capacitor consists of two parallel metal plates separated by a small distance as shown in Figure 1.52 (a).

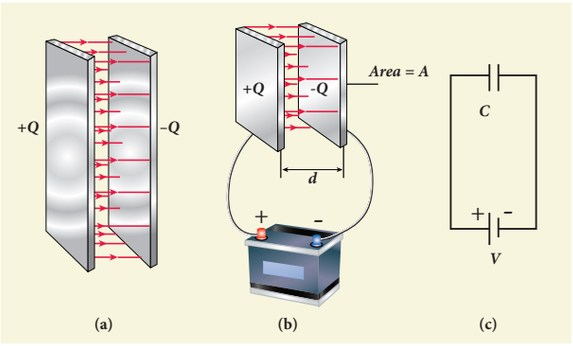


Figure 1.52 (a) Parallel plate capacitor (b) Capacitor connected with a battery (c) Symbolic representation of capacitor.

When a capacitor is connected to a battery of potential difference V, the electrons are transferred from one plate to the other plate by battery so that one plate becomes negatively charged with a charge of –Q and the other plate positively charged with +Q. The potential difference between the plates is equivalent to the battery's terminal voltage. This is shown in Figure 1.52 (b). If the battery voltage is increased, the amount of charges stored in the plates also increase. In general, the charge stored in the capacitor is proportional to the potential difference between the plates.

$$Q \propto V$$

so that $Q = CV$

where the C is the proportionality constant called capacitance. **The capacitance C of a capacitor is defined as the ratio of the magnitude of charge on either of the conductor plates to the potential difference existing between them.**

$$C = \frac{Q}{V} \qquad (1.81)$$

The SI unit of capacitance is *coulomb per volt* or *farad (*F*)* in honor of Michael Faraday. Farad is a larger unit of capacitance. In practice, capacitors are available in the range of microfarad ($1\mu F = 10^{-6}$ F) to picofarad ($1pF = 10^{-12}$ F). A capacitor is represented by the symbol ⊣⊢ or ⊣(. Note that the total charge stored in the capacitor is zero ($Q - Q = 0$). When we say the capacitor stores charges, it means the amount of charge that can be stored in any one of the plates.

Nowadays there are capacitors available in various shapes (cylindrical, disk) and types (tantalum, ceramic and electrolytic), as shown in Figure 1.53. These capacitors are extensively used in various kinds of electronic circuits.

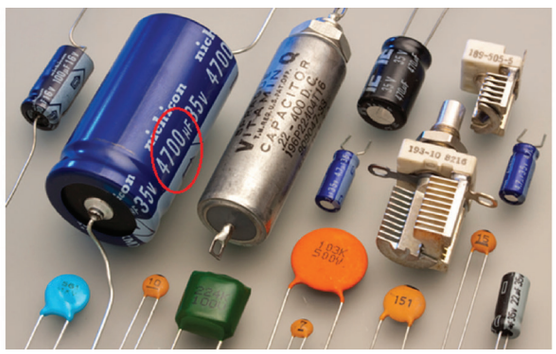


Figure 1.53 Various types of capacitors

Capacitance of a parallel plate capacitor

Consider a capacitor with two parallel plates each of cross-sectional area *A* and separated by a distance *d* as shown in Figure 1.54.

The electric field between two infinite parallel plates is uniform and is given by $E = \frac{\sigma}{\epsilon_0}$ where σ is the surface charge density on either plates $\left(\sigma = \frac{Q}{A}\right)$. If the separation distance *d* is very much smaller than the size of the plate ($d^2 << A$), then the above result can be used even for finite–sized parallel plate capacitor.

12th Physics_Vol-1_Unit 1_English Medium.indd 54 08-12-2022 11:03:57